

Multiple Choice

1. **Answer:** D

Options: A) Guttation is the outcome of rapid cellular respiration in plant tissues.; B) Guttation is a result of photosynthesis.; C) Guttation is caused by external weather conditions.; D) Guttation is explained by the effect of root pressure.; E) Guttation is a byproduct of the plant's defense mechanisms.

Note: Correct option: D - Guttation is explained by the effect of root pressure.

2. **Answer:** B

Options: A) The absence of a Y chromosome determines maleness in mammals; B) The presence of a Y chromosome usually determines maleness in mammals; C) The ratio of two X chromosomes to one Y chromosome determines maleness; D) The presence of an X chromosome determines maleness in mammals; E) The presence of an X chromosome in the absence of a Y chromosome determines maleness

Note: Correct option: B - The presence of a Y chromosome usually determines maleness in mammals

3. **Answer:** A

Options: A) The stomach has a high permeability to hydrogen ions; B) The presence of beneficial bacteria that neutralize stomach acid; C) The stomach lining is replaced every month; D) The stomach lining secretes digestive enzymes only when food is present; E) The production of bile in the stomach neutralizes the digestive acids

Note: Correct option: A - The stomach has a high permeability to hydrogen ions

4. **Answer:** A

Options: A) The zygote will not divide into two embryos, producing twins.; B) The sperm will not be able to fertilize the egg.; C) The sperm will fertilize the ovum.; D) The sperm will fertilize multiple eggs.; E) The zygote will begin to divide normally.

Note: Correct option: A - The zygote will not divide into two embryos, producing twins.

5. **Answer:** A

Options: A) It attracts predators away from the nestlings.; B) It aids in temperature regulation of the nest by reducing insulation.; C) It assists in the hatching of other eggs.; D) It is a ritualistic behavior without any particular significance.; E) It signals to other gulls that the nesting site is occupied.

Note: Correct option: A - It attracts predators away from the nestlings.

True / False

6. **Answer:** False

Options: A) True; B) False

Note: The metastasis rate of station 9 lymph nodes was significantly lower than other mediastinal stations in lung cancer patients. The metastasis status of station 9 had no significant influence on tumor staging or prognosis. Routine dissection of station 9 lymph nodes may not be necessary, especially in patients with a low T stage, upper or middle lobe tumors, or without intrapulmonary lymph node metastasis.

7. **Answer:** True

Options: A) True; B) False

Note: Our data, derived from patients with coronary artery disease, support the hypothesis regarding a possible preventive effect of bezafibrate on the development of colon cancer.

8. **Answer:** False

Options: A) True; B) False

Note: There is a knowledge deficit regarding the SU treatment goal among gout patients receiving ULT, despite generally high levels of other gout-specific knowledge. SU goal information may be an important and underutilized concept among providers treating gout patients.

9. **Answer:** False

Options: A) True; B) False

Note: Results of this survey highlight the wide variety of practice patterns in the US for handling surgical margins in breast-conservation treatment. This issue remains controversial, with no prevailing standard of care. Consequently, additional study is needed in the modern era of multimodality treatment to examine the minimal amount of surgical treatment necessary, in conjunction with chemotherapy and radiation, to attain adequate local control rates in breast-conservation treatment.

10. **Answer:** False

Options: A) True; B) False

Note: This study did not demonstrate preventive effects of family meetings on the mental health of family caregivers. Further research should determine whether this intervention might be more beneficial if provided in a more concentrated dose, when applied for therapeutic purposes or targeted towards subgroups of caregivers.

Long Form Response

11. **Answer:** An infected mother can transmit syphilis by placental transfer to the fetus during the first four months of pregnancy or as the fetus passes through the infected vagina during birth.. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

12. **Answer:** A decreased salt intake causes less water to be retained, decreasing blood volume, and in turn lowering blood pressure.. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.